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Blockchain Experiment 9

AIM: Implement the Private Ethereum Blockchain using Geth

THEORY:

Q1: What is Geth?

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go

programming language. It allows users to run an Ethereum node, mine Ether, and interact with

the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Q2: Significance of a Private Ethereum Network

Setting up a Private Ethereum Network involves creating a customized blockchain environment

isolated from the public Ethereum network. It enables developers to test smart contracts,

consensus mechanisms, and network behavior without incurring gas costs or depending on

public validators.

Q3: Steps for creating a Private Ethereum Network

Steps Involved:

● **Choosing a Network ID:**

Each private blockchain must have a unique network ID to differentiate it from public or

other private networks.

● **Choosing a Consensus Algorithm:**

Ethereum supports different consensus mechanisms such as Proof of Work (PoW) and

Proof of Authority (PoA). In private networks, Clique (PoA) is often used for faster

block confirmation.

● **Creating a Genesis Block:**

The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.

● **Initializing the Geth Database:**

The Geth init command initializes the blockchain using the genesis.json file, preparing the data directory for node operation.

● **Setting up Networking:**

Nodes communicate using enodes (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.

● **Running the Member Nodes:**

Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.

● **Running a Signer (in Clique):**

In the Clique consensus, designated signers validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

NOTE: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

PROCEDURE

Step 1: Install Geth

wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz

tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz

mkdir -p ~/bin

cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113

export PATH="$HOME/bin:$PATH"

geth113 version

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_ria_chaudhari@cloudshell:~$ wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version
--2026-04-08 17:25:19--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28248613 (27M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1'

geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1       100%[===================================================================================================>]  26.94M  4.86MB/s    in 8.1s

2026-04-08 17:25:28 (3.31 MB/s) - 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1' saved [28248613/28248613]

geth-linux-amd64-1.13.14-2bd6bd01/
geth-linux-amd64-1.13.14-2bd6bd01/COPYING
geth-linux-amd64-1.13.14-2bd6bd01/geth
Geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8561dd7fc21b631f4a34ac16627a1
Git Commit Date: 20240227
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=/home/g2022_ria_chaudhari/gopath:/google/gopath
GOROOT=
```

Step 2: Create folders and accounts for both nodes

```
g2022_ria_chaudhari@cloudshell:~$ mkdir -p ~/eth-private/node1 ~/eth-private/node2
g2022_ria_chaudhari@cloudshell:~$ geth113 --datadir ~/eth-private/node1 account new
INFO [04-08|17:27:46.537] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:27:46.540] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878
Path of the secret key file: /home/g2022_ria_chaudhari/eth-private/node1/keystore/UTC--2026-04-08T17-27-52.865021965Z--c8fe5b129bb5ee23ec4bdb4cab363ed92a2f9878

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

```
g2022_ria_chaudhari@cloudshell:~$ geth113 --datadir ~/eth-private/node2 account new
INFO [04-08|17:28:38.211] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:28:38.214] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x41911D883221851DD8C0Fe7C2a3E623D507bBb19
Path of the secret key file: /home/g2022_ria_chaudhari/eth-private/node2/keystore/UTC--2026-04-08T17-28-43.340511622Z--41911d883221851dd8c0fe7c2a3e623d507bbb19

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Step 3: Create password files

```
g2022_ria_chaudhari@cloudshell:~$ echo "1234" > ~/eth-private/node1/password.txt
echo "1234" > ~/eth-private/node2/password.txt
```

Step 4: Create genesis.json and verify it looks correct

NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE

"

NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE

"

```
cat > ~/eth-private/genesis.json << EOF

{

"config": {

"chainId": 1234,

"homesteadBlock": 0,

"eip150Block": 0,

"eip155Block": 0,

"eip158Block": 0,

"byzantiumBlock": 0,

"constantinopleBlock": 0,

"petersburgBlock":

0, "clique": {

"period": 5,

"epoch": 30000

}

},

"difficulty": "1",

"gasLimit": "8000000",

"extradata":

"0x0000000000000000000000000000000000000000000000000000000000000000${N
ODE1

}00000000000000000000000000000000000000000000000000000000000000000000
0000

0000000000000000000000000000000000000000000000000000000000000000",
```

"alloc": {

"0x${NODE1}": { "balance": "10000000000000000000" },

"0x${NODE2}": { "balance": "10000000000000000000" }

}

}

EOF

```
g2022_ria_chaudhari@cloudshell:~$ NODE1="C8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878"
NODE2="41911D883221851DD8C0Fe7C2a3E623D507bBb1"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
```

```
EOF
g2022_ria_chaudhari@cloudshell:~$ cat ~/eth-private/genesis.json
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000C8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878": { "balance": "10000000000000000000" },
    "0x41911D883221851DD8C0Fe7C2a3E623D507bBb1": { "balance": "10000000000000000000" }
  }
}
g2022_ria_chaudhari@cloudshell:~$
```

Step 5: Initialize both nodes

```
g2022_ria_chaudhari@cloudshell:~$ geth113 --datadir ~/eth-private/node1 init ~/eth-private/genesis.json
geth113 --datadir ~/eth-private/node2 init ~/eth-private/genesis.json
INFO [04-08|17:42:33.075] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:42:33.079] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:42:33.085] Set global gas cap                       cap=50,000,000
INFO [04-08|17:42:33.086] Initializing the KZG library             backend=gokzg
INFO [04-08|17:42:33.219] Defaulting to pebble as the backing database
INFO [04-08|17:42:33.219] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata cache=16.00MiB handles=16
INFO [04-08|17:42:33.300] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-08|17:42:33.300] State schema set to default              scheme=hash
INFO [04-08|17:42:33.301] Writing custom genesis block
INFO [04-08|17:42:33.304] Persisted trie from memory database      nodes=4 size=478.00B time=3.369394ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08|17:42:33.340] Successfully wrote genesis state         database=chaindata hash=38de1a..afc19d
INFO [04-08|17:42:33.340] Defaulting to pebble as the backing database
INFO [04-08|17:42:33.340] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-08|17:42:33.400] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [04-08|17:42:33.400] State schema set to default              scheme=hash
INFO [04-08|17:42:33.400] Writing custom genesis block
INFO [04-08|17:42:33.404] Persisted trie from memory database      nodes=4 size=478.00B time=3.273769ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08|17:42:33.438] Successfully wrote genesis state         database=lightchaindata hash=38de1a..afc19d
INFO [04-08|17:42:33.515] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:42:33.518] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:42:33.523] Set global gas cap                       cap=50,000,000
INFO [04-08|17:42:33.524] Initializing the KZG library             backend=gokzg
INFO [04-08|17:42:33.701] Defaulting to pebble as the backing database
INFO [04-08|17:42:33.701] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata cache=16.00MiB handles=16
INFO [04-08|17:42:33.758] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata/ancient/chain readonly=false
INFO [04-08|17:42:33.758] State schema set to default              scheme=hash
INFO [04-08|17:42:33.758] Writing custom genesis block
INFO [04-08|17:42:33.762] Persisted trie from memory database      nodes=4 size=478.00B time=3.381802ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08|17:42:33.798] Successfully wrote genesis state         database=chaindata hash=38de1a..afc19d
INFO [04-08|17:42:33.798] Defaulting to pebble as the backing database
INFO [04-08|17:42:33.798] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node2/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-08|17:42:33.853] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node2/geth/lightchaindata/ancient/chain readonly=false
INFO [04-08|17:42:33.853] State schema set to default              scheme=hash
INFO [04-08|17:42:33.853] Writing custom genesis block
INFO [04-08|17:42:33.857] Persisted trie from memory database      nodes=4 size=478.00B time=3.539219ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08|17:42:33.891] Successfully wrote genesis state         database=lightchaindata hash=38de1a..afc19d
g2022_ria_chaudhari@cloudshell:~$
```

Step 6: Run Node 1 (signer)

```
g2022_ria_chaudhari@cloudshell:~$ geth113 --datadir ~/eth-private/node1 \
  --networkid 1234 \
  --port 30303 \
  --http --http.port 8545 \
  --http.api personal,eth,net,web3,miner \
  --unlock 0x${NODE1} \
  --password ~/eth-private/node1/password.txt \
  --mine \
  --miner.etherbase 0x${NODE1} \
  --allow-insecure-unlock \
  console
INFO [04-08|17:44:52.561] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:44:52.564] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:44:52.568] Set global gas cap                       cap=50,000,000
INFO [04-08|17:44:52.569] Initializing the KZG library             backend=gokzg
INFO [04-08|17:44:52.700] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|17:44:52.701] Using pebble as the backing database
INFO [04-08|17:44:52.701] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [04-08|17:44:52.730] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-08|17:44:52.730] State scheme set to already existing     scheme=hash
INFO [04-08|17:44:52.732] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-08|17:44:52.732]
INFO [04-08|17:44:52.733] ------------------------------------------------------------------------------------------------------------------------------------------------------
INFO [04-08|17:44:52.733] Chain ID:  1234 (unknown)
INFO [04-08|17:44:52.733] Consensus: Clique (proof-of-authority)
INFO [04-08|17:44:52.733]
INFO [04-08|17:44:52.733] Pre-Merge hard forks (block based):
INFO [04-08|17:44:52.733]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-08|17:44:52.733]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [04-08|17:44:52.733]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08|17:44:52.733]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08|17:44:52.733]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [04-08|17:44:52.733]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.md)
INFO [04-08|17:44:52.733]  - Petersburg:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/petersburg.md)
INFO [04-08|17:44:52.733]  - Istanbul:                     #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/istanbul.md)
INFO [04-08|17:44:52.733]  - Berlin:                       #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/berlin.md)
INFO [04-08|17:44:52.733]  - London:                       #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/london.md)
```

```
instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
coinbase: 0xc8fe5b129bb5ee23ec4bdb4cab363ed92a2f9878
at block: 1 (Wed Apr 08 2026 17:44:54 GMT+0000 (UTC))
 datadir: /home/g2022_ria_chaudhari/eth-private/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-08|17:44:54.582] New local node record                    seq=1,775,670,194,395 id=ab0dd2ac241e15ee ip=34.21.175.36 udp=30303 tcp=30303
INFO [04-08|17:44:59.007] Successfully sealed new block             number=2 sealhash=163e73..f7277c hash=87483e..f8f261 elapsed=4.954s
INFO [04-08|17:44:59.008] Commit new sealing work                   number=3 sealhash=ce7107..eda36e txs=0 gas=0 fees=0 elapsed="395.747µs"
INFO [04-08|17:45:02.925] Looking for peers                         peercount=0 tried=108 static=0
INFO [04-08|17:45:04.007] Successfully sealed new block             number=3 sealhash=ce7107..eda36e hash=f92c48..6be3ea elapsed=4.999s
INFO [04-08|17:45:04.008] Commit new sealing work                   number=4 sealhash=346563..38e33e txs=0 gas=0 fees=0 elapsed="418.854µs"
INFO [04-08|17:45:09.008] Successfully sealed new block             number=4 sealhash=346563..38e33e hash=b736c7..b322b8 elapsed=4.999s
INFO [04-08|17:45:09.008] Commit new sealing work                   number=5 sealhash=4fa3a0..c18fe8 txs=0 gas=0 fees=0 elapsed="388.844µs"
INFO [04-08|17:45:12.929] Looking for peers                         peercount=0 tried=82  static=0
INFO [04-08|17:45:14.009] Successfully sealed new block             number=5 sealhash=4fa3a0..c18fe8 hash=2cd841..fe601a elapsed=5.000s
INFO [04-08|17:45:14.009] Commit new sealing work                   number=6 sealhash=a48aa5..fc52db txs=0 gas=0 fees=0 elapsed="336.108µs"
INFO [04-08|17:45:19.008] Successfully sealed new block             number=6 sealhash=a48aa5..fc52db hash=f436f2..354195 elapsed=4.999s
INFO [04-08|17:45:19.009] Commit new sealing work                   number=7 sealhash=02b9f2..f772ac txs=0 gas=0 fees=0 elapsed="497.656µs"
INFO [04-08|17:45:23.331] Looking for peers                         peercount=0 tried=26  static=0
INFO [04-08|17:45:24.008] Successfully sealed new block             number=7 sealhash=02b9f2..f772ac hash=5ca646..bfc786 elapsed=4.999s
INFO [04-08|17:45:24.009] Commit new sealing work                   number=8 sealhash=212487..bcdc3f txs=0 gas=0 fees=0 elapsed="366.968µs"
INFO [04-08|17:45:29.008] Successfully sealed new block             number=8 sealhash=212487..bcdc3f hash=b48c05..084ec5 elapsed=4.999s
INFO [04-08|17:45:29.009] Commit new sealing work                   number=9 sealhash=b6c8db..a8a196 txs=0 gas=0 fees=0 elapsed="337.61µs"
INFO [04-08|17:45:33.555] Looking for peers                         peercount=0 tried=91  static=0
```

Step 7: Open new terminal tab and run Node 2 (member)

```
g2022_ria_chaudhari@cloudshell:~$ geth113 --datadir ~/eth-private/node2 \
  --networkid 1234 \
  --port 30304 \
  --http --http.port 8546 \
  --authrpc.port 8552 \
  --http.api personal,eth,net,web3 \
  --unlock 0x${NODE2} \
  --password ~/eth-private/node2/password.txt \
  --allow-insecure-unlock \
  console
INFO [04-08|17:48:52.981] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:48:52.985] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:48:52.989] Set global gas cap                       cap=50,000,000
INFO [04-08|17:48:52.989] Initializing the KZG library             backend=gokzg
INFO [04-08|17:48:53.126] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|17:48:53.126] Using pebble as the backing database
INFO [04-08|17:48:53.127] Allocated cache and file handles         database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [04-08|17:48:53.151] Opened ancient database                  database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata/ancient/chain readonly=false
INFO [04-08|17:48:53.151] State scheme set to already existing     scheme=hash
INFO [04-08|17:48:53.153] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] ----------------------------------------------------------------------------------------------------------------------------------
INFO [04-08|17:48:53.154] Chain ID:  1234 (unknown)
INFO [04-08|17:48:53.154] Consensus: Clique (proof-of-authority)
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] Pre-Merge hard forks (block based):
INFO [04-08|17:48:53.154]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-08|17:48:53.154]  - Tangerine Whistle  (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [04-08|17:48:53.154]  - Spurious Dragon/1  (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08|17:48:53.154]  - Spurious Dragon/2  (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08|17:48:53.154]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [04-08|17:48:53.154]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.md)
INFO [04-08|17:48:53.154]  - Petersburg:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/petersburg.md)
INFO [04-08|17:48:53.154]  - Istanbul:                     #<nil>  (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/istanbul.md)
INFO [04-08|17:48:53.154]  - Berlin:                       #<nil>  (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/berlin.md)
INFO [04-08|17:48:53.154]  - London:                       #<nil>  (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/london.md)
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] The Merge is not yet available for this network!
```

```
68034ec63cf3238@127.0.0.1:30304
INFO [04-08|17:48:53.244] WebSocket enabled                        url=ws://127.0.0.1:8552
INFO [04-08|17:48:53.244] HTTP server started                      endpoint=127.0.0.1:8552 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-08|17:48:54.532] Unlocked account                         address=0x41911D883221851DD8C0Fe7C2a3E623D507bBb19
WARN [04-08|17:48:54.584] Served eth_coinbase                      reqid=3 duration="52.151µs" err="etherbase must be explicitly specified"
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
 datadir: /home/g2022_ria_chaudhari/eth-private/node2
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-08|17:48:55.861] New local node record                    seq=1,775,670,478,928 id=3c0cea706c0e875d ip=34.21.175.36 udp=30304 tcp=30304
INFO [04-08|17:49:03.289] Looking for peers                        peercount=1 tried=60 static=0
INFO [04-08|17:49:13.303] Looking for peers                        peercount=0 tried=75 static=0
INFO [04-08|17:49:24.592] Looking for peers                        peercount=0 tried=39 static=0
INFO [04-08|17:49:34.781] Looking for peers                        peercount=0 tried=67 static=0
```

Step 8: Connect the two nodes In Node 1 console, get its enode

```
> admin.nodeInfo.enode
"enode://37a3db1c1e34eecb3611941ca5b7001168ac8eec28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb@34.21.175.36:30303"
```

Now, switch to Node 2 console and paste the Node 1 enode

```
> admin.addPeer("enode://37a3db1c1e34eecb3611941ca5b7001168ac8eec28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb@34.21.175.36:30303")
true
```

In Node 2 console, check they are connected

```
> admin.peers INFO [04-08|18:01:18.017] Imported new chain segment               number=196 hash=9ac569..83685f blocks=1   txs=0 mgas=0.000 elapsed=7.746ms   mgasps=0.000 triedirt

[{
    caps: ["eth/68", "snap/1"],
    enode: "enode://37a3db1c1e34eecb3611941ca5b7001168ac8eec28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb@127.0.0.1:30303",
    id: "ab0dd2ac241e15eec929ff71bbb9d19f854213e77207f9ee0b74c4911801aef7",
    name: "Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6",
    network: {
      inbound: false,
      localAddress: "127.0.0.1:60510",
      remoteAddress: "127.0.0.1:30303",
      static: true,
      trusted: false
    },
    protocols: {
      eth: {
        version: 68
      },
      snap: {
        version: 1
      }
    }
]]
```

Step 9: Send a transaction In Node 1 console, send transaction to Node 2

```
> eth.sendTransaction({
......   from: "0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878",
......   to: "0x41911D883221851DD8C0Fe7C2a3E623D507bBb19",
......   value: "0xde0b6b3a7640000"
...... })
INFO [04-08|18:03:16.769] Setting new local account                address=0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878
INFO [04-08|18:03:16.770] Submitted transaction                    hash=0x6e201621cef61dc5e3c3f114cbd03055c23b1f083ebf98411f8c91cb48e47265 from=0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878 nonce=
0 recipient=0x41911D883221851DD8C0Fe7C2a3E623D507bBb19 value=1,000,000,000,000,000,000
"0x6e201621cef61dc5e3c3f114cbd03055c23b1f083ebf98411f8c91cb48e47265"
```

and check the balance of Node 2

```
> web3.fromWei(
...    eth.getBalance("0x41911D883221851DD8C0Fe7C2a3E623D507bBb19"),
...    "ether"
... )
101
```

CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing, compiling, deploying, and testing smart contracts on the Ethereum blockchain.